

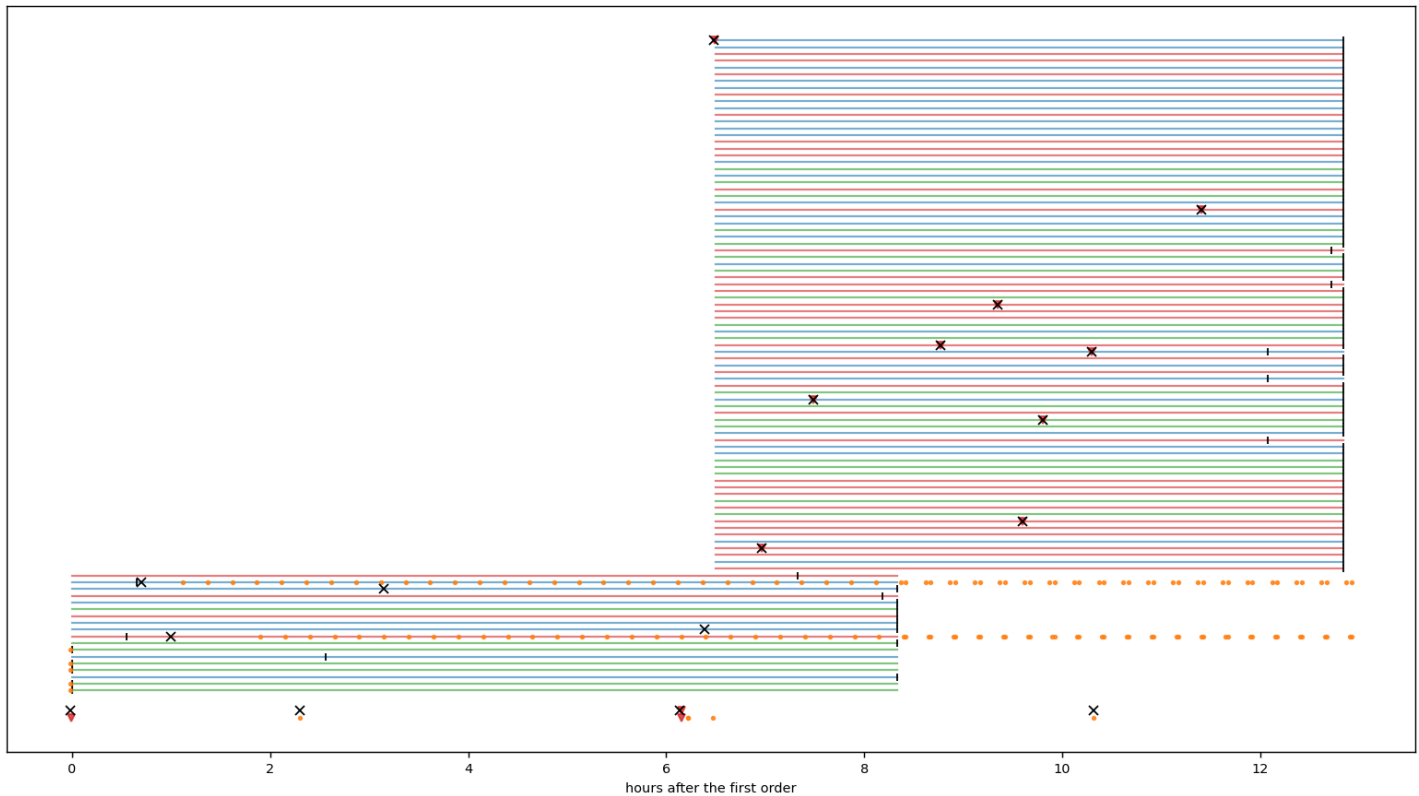
Systematic Execution Operations Stack

Question. A systematic multi-asset fund sends hundreds of orders a day through brokers' algorithms. What has to be checked before, during and after the day for the book to be right, how quickly can a monitor catch what goes wrong, and what does execution cost? This stack answers with measured numbers: a fault-injection harness for the live monitor, a scored three-way reconciliation, a corporate-actions engine checked against twenty years of vendor data, exchange and contract calendars checked against the data, and transaction cost analysis with a broker wheel.

Method. Python package (xops) with its own FIX 4.4 engine (session layer with heartbeats, test requests, resend and gap fill, sequence persistence; NewOrderSingle, cancel, cancel-replace, ExecutionReport, OrderCancelReject) that runs identically on real sockets and on the simulator's virtual clock. Three simple systematic strategies on free daily data generate the flow; portfolio-to-orders nets across strategies, rounds, applies the restricted list, locates and participation caps and schedules by exchange session; three simulated brokers run VWAP, TWAP, POV, implementation-shortfall and close algorithms against a reduced-form intraday market with square-root impact and report every fill through FIX and a drop-copy session. The monitor keeps an order state machine per order and evaluates its rules every fifteen seconds. The end of day reconciles the blotter, the drop copy, a simulated prime-broker file with seeded breaks, and positions, in SQL on DuckDB. Fault days alternate with clean days so that detection and false-alert rates are both measured.

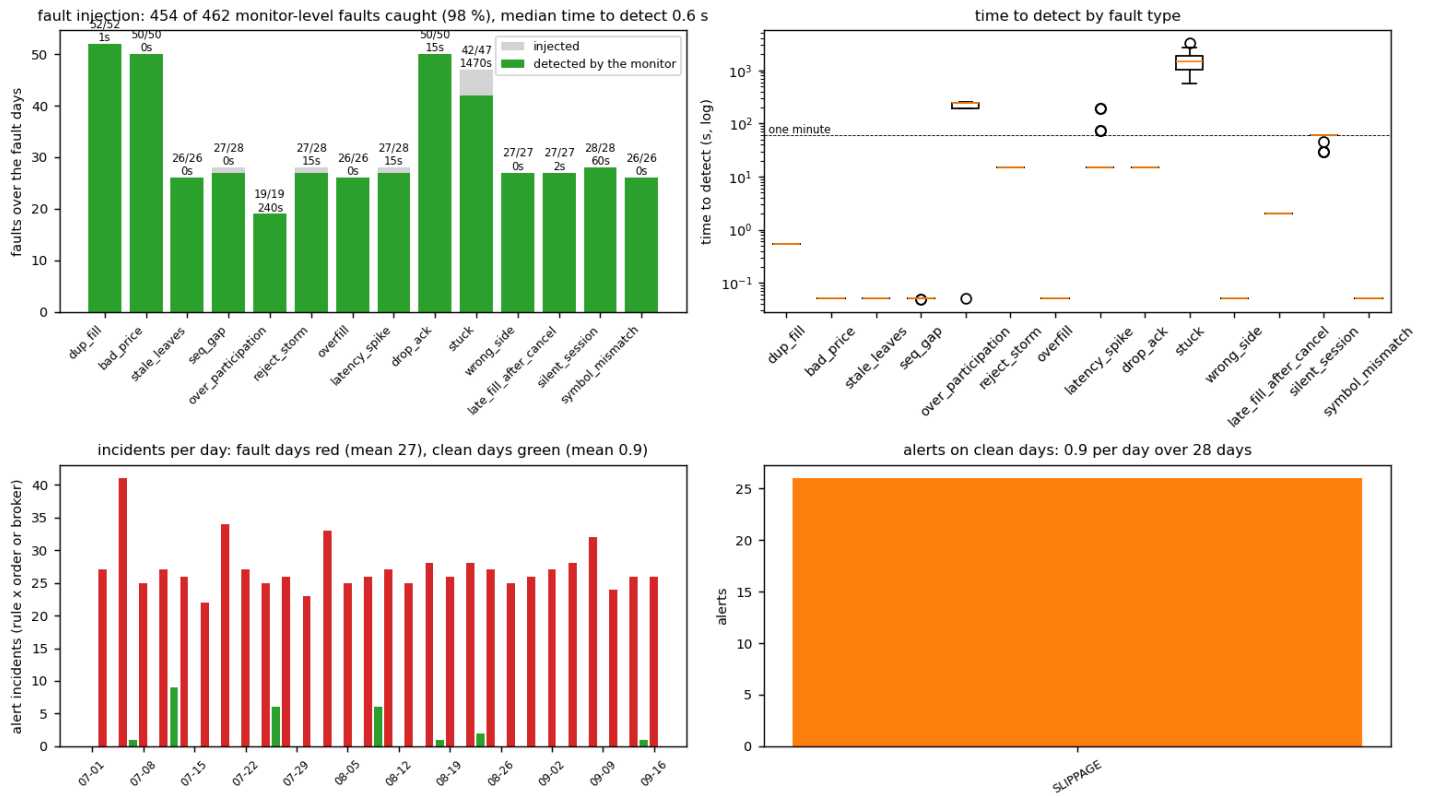
Caveats. Brokers, fills and the prime broker are simulated; the price data, corporate actions, calendars and the vendor's inconsistencies are real. The market model is reduced-form, so the transaction-cost levels are stylised; the monitor, reconciliation and corporate-action results do not depend on it.

2026-07-08: 97 orders (colour = broker, tick = filled fraction), x = injected fault, red = high alert, orange = other alert; session-level alerts below the axis

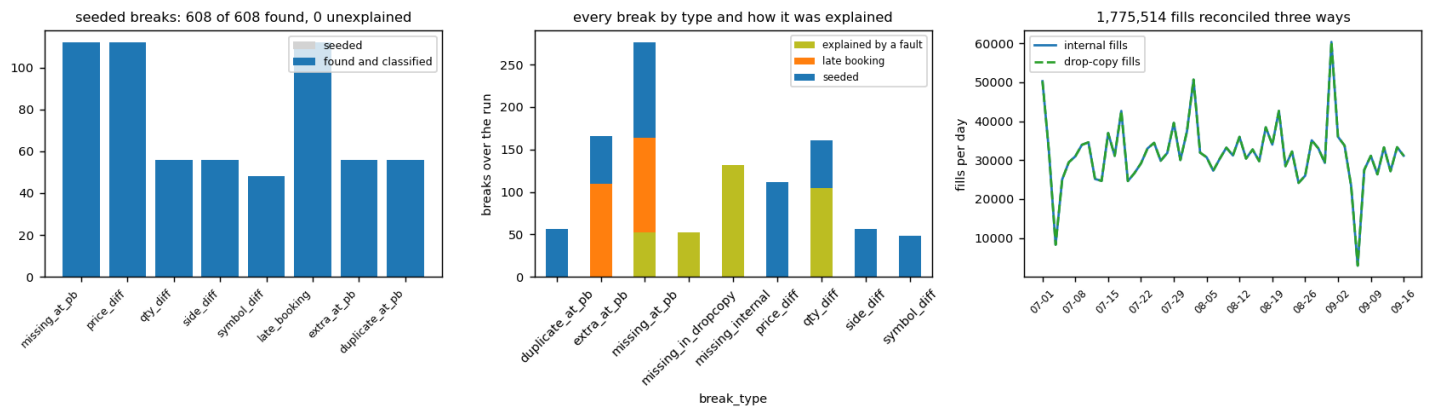


One fault day: every order on a timeline, coloured by broker, with injected faults (x) and the monitor's alerts.

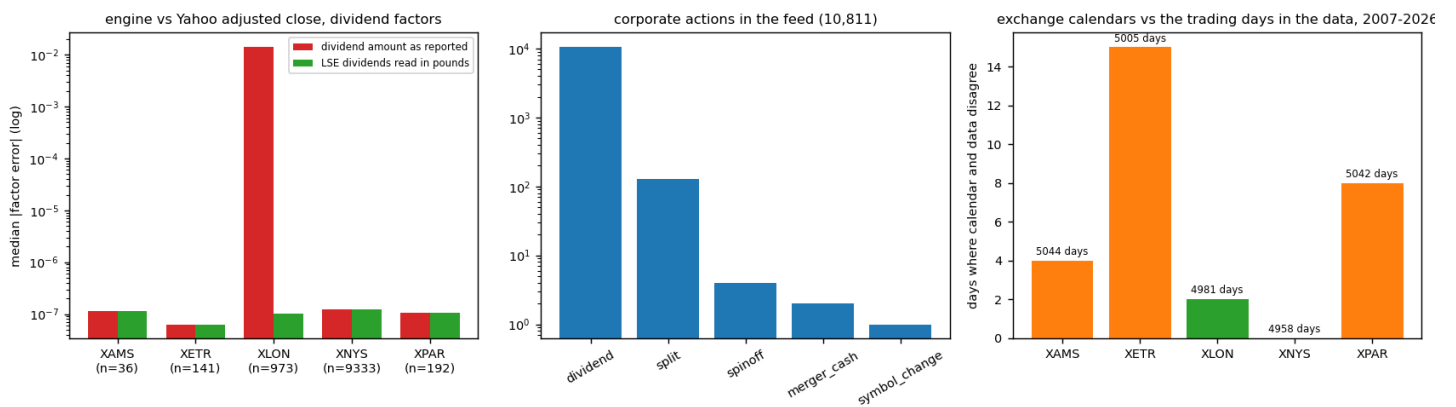
systematic-execution-ops - portfolio-to-orders, FIX order lifecycle, live monitor, reconciliation, corporate actions



Fault injection: detection by fault type and time to detect, incidents per day on fault and clean days, alerts on clean days by rule.

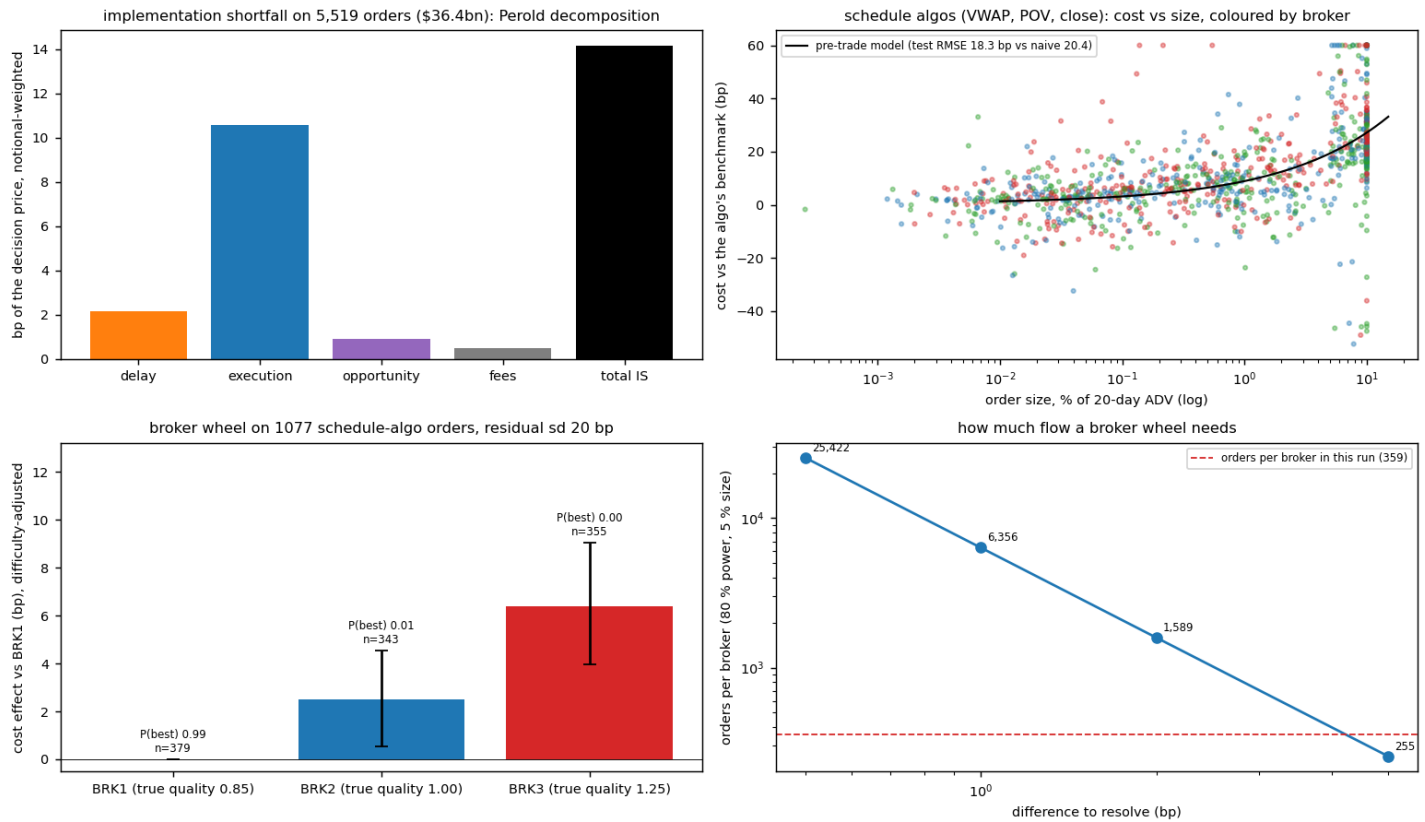


End-of-day reconciliation: seeded breaks found by type, every break by how it was explained, fills reconciled per day.



Corporate actions and calendars: dividend factors against Yahoo's adjusted close (and the LSE unit inconsistency), events in the feed, calendar-versus-data mismatches.

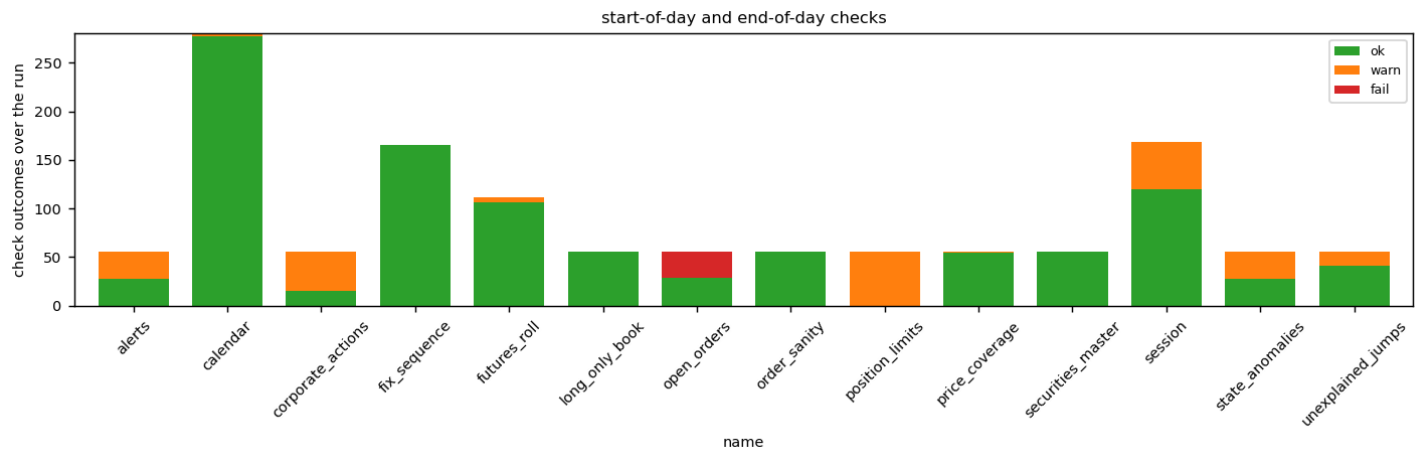
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Transaction costs: implementation-shortfall decomposition, cost against size with the pre-trade model, broker wheel effects with intervals, and the flow a wheel needs.



Portfolio-to-orders: netting per day, algorithm mix, acknowledgement latency by broker, terminal states.



Start-of-day and end-of-day check outcomes over the run.

Generated by scripts/summarize.py from results/*.json.

The run: 2026-07-01 to 2026-09-16, 56 trading days (28 with injected faults, 28 clean), AUM \$2bn, brokers BR

Orders 5,663 (101 per day), fills booked 1,775,514, drop-copy fills 1,775,646; gross strategy deltas \$55.0bn of which \$1.67bn crossed internally and \$43.6bn sent; corporate actions applied on 173 position lines; runtime 14 minutes.

									date
2026-07-01	clean	135	50282	0	-	11/11	0	3/0	
2026-07-02	fault	99	31903	27	17/18	11/11	0	6/1	
2026-07-03	clean	17	8290	0	-	10/10	0	2/0	
2026-07-06	fault	115	25045	41	14/18	11/11	0	6/1	
2026-07-07	clean	84	29441	1	-	11/11	0	1/0	
2026-07-08	fault	97	30982	25	15/17	11/11	0	6/1	
2026-07-09	clean	101	33930	0	-	11/11	0	2/0	
2026-07-10	fault	108	34579	27	18/19	11/11	0	6/1	
2026-07-13	clean	108	25169	9	-	11/11	0	2/0	
2026-07-14	fault	75	24678	26	15/19	11/11	0	4/1	
2026-07-15	clean	101	36957	0	-	11/11	0	3/0	
2026-07-16	fault	96	31108	22	15/17	11/11	0	6/0	
2026-07-17	clean	113	42632	0	-	11/11	0	1/0	
2026-07-20	fault	123	24629	34	16/18	11/11	0	7/1	
2026-07-21	clean	86	26659	0	-	11/11	0	1/0	
2026-07-22	fault	94	29178	27	16/18	10/10	0	6/1	
2026-07-23	clean	95	33013	0	-	11/11	0	2/0	
2026-07-24	fault	104	34442	25	17/18	11/11	0	5/1	
2026-07-27	clean	132	29852	6	-	11/11	0	3/0	
2026-07-28	fault	100	31839	26	14/17	11/11	0	6/1	
2026-07-29	clean	109	39571	0	-	10/10	0	1/0	
2026-07-30	fault	92	30033	23	16/18	11/11	0	4/1	
2026-07-31	clean	101	37383	0	-	11/11	0	4/0	
2026-08-03	fault	165	50685	33	18/20	11/11	0	7/1	
2026-08-04	clean	93	31935	0	-	11/11	0	1/0	
2026-08-05	fault	98	30730	25	17/19	11/11	0	4/1	
2026-08-06	clean	86	27291	0	-	10/10	0	2/0	
2026-08-07	fault	100	30373	26	16/18	11/11	0	6/1	
2026-08-10	clean	131	33205	6	-	11/11	0	2/0	
2026-08-11	fault	96	31194	27	17/20	11/11	0	6/1	
2026-08-12	clean	102	35978	0	-	10/10	0	2/0	
2026-08-13	fault	96	30399	25	16/18	10/10	0	6/1	
2026-08-14	clean	96	32746	0	-	11/11	0	2/0	
2026-08-17	fault	124	29723	28	16/18	11/11	0	5/1	
2026-08-18	clean	106	38428	1	-	11/11	0	1/0	
2026-08-19	fault	101	33987	26	17/19	11/11	0	5/1	
2026-08-20	clean	112	42645	0	-	11/11	0	3/0	
2026-08-21	fault	91	28452	28	16/19	10/10	0	7/1	
2026-08-24	clean	130	32223	2	-	11/11	0	2/0	
2026-08-25	fault	83	24151	27	17/18	11/11	0	6/1	
2026-08-26	clean	86	26010	0	-	11/11	0	3/0	
2026-08-27	fault	109	35101	25	17/19	11/11	0	6/1	
2026-08-28	clean	98	33054	0	-	11/11	0	4/0	
2026-08-31	fault	123	29302	26	16/19	11/11	0	7/1	
2026-09-01	clean	158	60357	0	-	11/11	0	3/0	
2026-09-02	fault	110	35995	27	17/19	11/11	0	6/1	
2026-09-03	clean	99	33746	0	-	11/11	0	2/0	
2026-09-04	fault	86	23391	28	18/20	11/11	0	6/1	
2026-09-07	clean	8	2910	0	-	10/10	0	2/0	
2026-09-08	fault	113	27512	32	16/18	11/11	0	7/1	
2026-09-09	clean	87	31116	0	-	11/11	0	2/0	
2026-09-10	fault	92	26363	24	15/17	11/11	0	6/1	

2026-09-11	clean	99	33284	0	-	11/11	0	2/0
2026-09-14	fault	125	27170	26	15/17	11/11	0	7/1
2026-09-15	clean	89	33325	1	-	11/11	0	3/0
2026-09-16	fault	86	31138	26	17/19	11/11	0	7/1

Live monitor: fault injection

Over the 28 fault days 514 faults were injected into the brokers and the transport; the monitor caught 454 (88 %) with a median time to detect of 1 s (p90 255 s). The two that are not the monitor's job (dropcopy_missing) are found by the end-of-day reconciliation. On the 28 clean days the monitor raised 0.9 alerts per day (0.9 incidents).

						fault
dup_fill	52	52	1	1	STATE_ANOMALY duplicate	ExecID
bad_price	50	50	0	0	PRICE_COLLAR	
stale_leaves	26	26	0	0	STATE_ANOMALY LeavesQty	
seq_gap	28	27	0	0	SEQ_GAP	
dropcopy_missing	52	0	-	-	(EOD reconciliation)	
over_participation	19	19	240	255	PARTICIPATION / AHEAD_SCHEDULE	
reject_storm	28	27	15	15	REJECT_RATE	
overfill	26	26	0	0	STATE_ANOMALY overfill	
latency_spike	28	27	15	123	LATENCY	
drop_ack	50	50	15	15	UNACKED	
stuck	47	42	1470	2100	STALLED / BEHIND_SCHEDULE / LEA	
wrong_side	27	27	0	0	STATE_ANOMALY side	
late_fill_after_cancel	27	27	2	2	STATE_ANOMALY fill after Canceled	
silent_session	28	28	60	60	SESSION_SILENT / SESSION_LOST	
symbol_mismatch	26	26	0	0	STATE_ANOMALY symbol	

Alerts on clean days by rule: SLIPPAGE 26.

End-of-day reconciliation

1,775,514 internal fills reconciled against the drop copy and the prime broker file over 56 days. Seeded breaks: 608 of 608 found and classified; 0 breaks unexplained (a clean day reconciles to zero).

			break ty
missing_at_pb	112	112	
price_diff	112	112	
qty_diff	56	56	
side_diff	56	56	
symbol_diff	48	48	
late_booking	112	112	
extra_at_pb	56	56	
duplicate_at_pb	56	56	

Every break raised over the run, by type and label (label = the seeded break it matched, the fault or rejected report that explains it, late_booking / late_booking_resolved for the two-day pair, or blank when unexplained):

			break ty
duplicate_at_pb	duplicate_at_pb	56	
extra_at_pb	extra_at_pb	56	
extra_at_pb	late_booking_resolved	110	
missing_at_pb	fault:dropcopy_missing	52	
missing_at_pb	late_booking	112	
missing_at_pb	missing_at_pb	112	
missing_in_dropcopy	fault:dropcopy_missing	52	
missing_internal	fault:late_fill_after_cancel	27	
missing_internal	fault:overfill	26	
missing_internal	fault:stale_leaves	26	
missing_internal	fault:symbol_mismatch	26	
missing_internal	fault:wrong_side	27	
price_diff	price_diff	112	
qty_diff	qty_diff	56	

qty_diff	resync_after_rejected_report	105
side_diff	side_diff	56
symbol_diff	symbol_diff	48

Transaction cost analysis

5,519 executed orders, \$36.4bn. Implementation shortfall 14.1 bp of the decision price = delay 2.2 + execution 10.6 + opportunity 0.9 + fees 0.5 (Perold; exact by construction). Execution vs arrival 11.8 bp, vs interval VWAP 13.1 bp, fill rate 83 %.

CLOSE	533	8.1	21.8	19.7	99 %	algo
IS	4442	10.9	10.9	11.5	99 %	
POV	348	26.0	29.3	26.0	48 %	
VWAP	196	8.1	-19.7	8.1	97 %	

difficulty (0 <1% ADV, 1 1-5%, 2 >5%)	orders	vs arrival (bp)	vs VWAP (bp)	pre-trade estimate (bp)
0	4229	8.0	9.4	6.2
1	942	17.0	19.5	15.2
2	348	29.3	26.0	34.5

Broker wheel (1077 schedule-algo orders: VWAP and POV against the interval VWAP, close orders against the close; cost regressed

broker	orders	raw cost vs benchmark (bp)	impact vs BRK1 (bp)	95 % CI	P(best)	by algo (bp)
BRK1	379	13.6	+0.0	[+0.0, +0.0]	0.99	CLOSE 6.2, POV 20.9, VWAP 8.6
BRK2	343	14.6	+2.5	[+0.5, +4.5]	0.01	CLOSE 6.5, POV 26.4, VWAP 8.0
BRK3	355	18.9	+6.4	[+4.0, +9.1]	0.00	CLOSE 11.3, POV 32.5, VWAP 7.8

Orders per broker needed to resolve a difference at 80 % power: 0.5bp: 25,422, 1.0bp: 6,356, 2.0bp: 1,589, 5.0bp: 255. Ranking by estimated effect: BRK1 < BRK2 < BRK3; the simulator's true quality order is BRK1 < BRK2 < BRK3 (impact multipliers 0.85, 1.00, 1.25).

On all 5519 orders including the implementation-shortfall ones measured against the interval VWAP (residual sd 49 bp) the effects are BRK1 +0.0 [+0.0, +0.0], BRK2 +3.3 [-1.5, +7.8], BRK3 +5.3 [+0.6, +9.8]; resolving 2 bp would take 9,227 orders per broker. The benchmark has to fit the algo before a wheel can rank anyone.

Pre-trade model on the schedule-algo orders: cost = 0.8 + 0.55 x sigma_bp x sqrt(Q/ADV) + -0.19 x spread_bp, fitted on 686 orders before 2026-08-21, tested on 391: RMSE 18.3 bp against 20.4 for the training mean, out-of-sample correlation 0.45.

Corporate actions

Feed: dividend 10,675, split 129, spinoff 4, merger_cash 2, symbol_change 1. Dividend factors checked against Yahoo's adjusted close: 10,675 events; as reported 9,699 within 5e-4 (median relative error 1.4e-07); with LSE dividends read in pounds 10,672 within tolerance (median 1.2e-07, max 1.4e-03); 0 adjustment days without an event in the feed.

XLON	973	0	1.4e-02	973	1.0e-07	exchange
XNYS	9333	9333	1.2e-07	9333	1.2e-07	
XPAR	192	190	1.1e-07	190	1.1e-07	
XETR	141	140	6.4e-08	140	6.4e-08	
XAMS	36	36	1.1e-07	36	1.1e-07	

Yahoo's close series is already split-adjusted, so dividends are the only events its adjusted close can check; LSE prices are in pence while Yahoo adjusts with the dividend in pounds, a factor-100 vendor inconsistency that the check finds

Exchange calendars vs the trading days in the data

exchange	symbols	period	calendar days	data days	calendar open but no data	data on a calendar-closed day
XAMS	1	2007-01-01..2026-09-17	5046	5044	2010-12-31, 2015-12-31, 2020-12-31	2007-01-01
XETR	7	2007-01-02..2026-09-17	5018	5005	2007-05-28, 2011-10-03, 2015-10-26	2013-1-1
XLON	20	2007-01-02..2026-09-17	4981	4981	2012-05-28	2011-04-29
XNYS	157	2007-01-03..2026-09-17	4958	4958	-	-
XPAR	6	2007-01-02..2026-09-17	5046	5042	2010-12-24, 2010-12-31, 2014-12-31, 2019-12-25	2012-11-25

Fixed-income and derivatives calendars

contract	first notice	last trade
ZNZ25	2025-11-28	2025-12-19
ZNH26	2026-02-27	2026-03-20
ZNM26	2026-05-29	2026-06-18

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ZNU26	2026-08-31	2026-09-21
ZNZ26	2026-11-30	2026-12-21
ZTZ25	2025-11-28	2025-12-31
ZBZ25	2025-11-28	2025-12-19
SR3Z25	-	2026-03-17
SR3H26	-	2026-06-16
ESZ25	-	2025-12-19
ESH26	-	2026-03-20
ESU26	-	2026-09-18
6EZ25	-	2025-12-15
GCZ25	2025-11-28	2025-12-29
CLX26	2026-10-21	2026-10-20
CLZ26	2026-11-23	2026-11-20
NGX26	-	2026-10-28

Front contracts today: ZN ZNZ26, ES ESU26, SR3 SR3U26, CL CLV26. CDX rolls ['2026-03-20', '2026-09-21']; monthly option expiries 2026: 2026-01-16, 2026-02-20, 2026-03-20, 2026-04-17, 2026-05-15, 2026-06-18, 2026-07-17, 2026-08-21, 2026-09-18, 2026-10-16, 2026-11-20, 2026-12-18. TBA (SIFMA) 2026-01 A: notification 2026-01-12, settlement 2026-01-14; 2026-01 B: notification 2026-01-15, settlement 2026-01-20; 2026-01 C: notification 2026-01-20, settlement 2026-01-22; 2026-01 D: notification 2026-01-23, settlement 2026-01-27. TIPS reference CPI today 333.934 from 33 months of CPI-U.

Checks

phase	check	ok	warn	fail
EOD	alerts	28	28	0
EOD	long_only_book	56	0	0
EOD	open_orders	29	0	27
EOD	session	120	48	0
EOD	state_anomalies	28	28	0
SOD	calendar	277	3	0
SOD	corporate_actions	15	41	0
SOD	fix_sequence	165	0	0
SOD	futures_roll	106	6	0
SOD	order_sanity	56	0	0
SOD	position_limits	0	56	0
SOD	price_coverage	54	2	0
SOD	securities_master	56	0	0
SOD	unexplained_jumps	41	15	0